IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

1 Methods

SpaceX API and Wikipedia collection, wrangling, SQL, mapping and modeling.

2 Results

Landing outcomes by site, orbit and year. Folium maps and Dash views put the patterns in context.

3 Prediction and conclusions

Four classifiers, an 18-flight test, and a check on how the models handle unsuccessful outcomes.

Executive summary

60 of 90 flights

Two thirds of the Falcon 9 records had a positive first-stage outcome, including five controlled ocean landings.

10 of 13 at Kennedy

KSC LC-39A had the highest within-site success rate in the 56-flight dashboard sample (76.9%).

15 of 18 test predictions

All four classifiers tied at 83.3%. In the verification run, every model missed three of the six class 0 outcomes.

Can launch records predict a first-stage landing?

The business question

Recovering a booster makes reuse possible. A competitor assessing a bid would want to know how reliably recovery can be anticipated.

The analytical question

How does the recorded outcome vary with payload, destination orbit, launch site and flight history?

What the data measure

The target is the course-defined first-stage outcome. Estimating refurbishment cost or actual savings would require additional data.

Section 1

Methodology



Methodology

Collection and preparation

Retrieve historical launch records and parse Wikipedia tables. Use the course extracts to define a one-row-per-flight modeling table.

Exploration

Plot landing outcomes, query launch records in SQLite, and explore individual sites with Folium and Dash.

Evaluation

Tune four classifiers using cross-validation on the training set. Evaluate them on the same 18 held-out flights.

Three course extracts answer different questions

90 flights for EDA and prediction

The labeled Falcon 9 file has three launch-site labels. Its 80-column feature matrix supports the classification experiment.

101 records for SQL

The SQL file records mission and landing outcomes separately and distinguishes four site labels.

56 flights for interactive views

The map and dashboard sample uses four site labels. Percentages from this sample have a different denominator from the modeling data.

SpaceX API collection

Request

The live launches endpoint returned HTTP 522 in the saved run. The archived course response returned HTTP 200.

Parse and enrich

The notebook normalizes the JSON and defines lookups for rockets, payloads, cores and launchpads. Some live lookups returned empty results.

Recover a usable dataset

The wrangling notebook reads IBM’s prepared dataset_part_1.csv. The labeled snapshot provides the 90 records analyzed here.

Wikipedia web scraping

Retrieve the archived page

The saved request returned HTTP 200. BeautifulSoup found the launch tables on the Falcon 9 and Falcon Heavy launch list.

Parse the table rows

Extract flight number, date and time, booster, site, payload mass, orbit and customer. Read mission and booster outcomes separately.

Keep the sources distinct

The notebook includes a CSV export for the scraped table. The classification labels come from the course’s Falcon 9 dataset.

How the landing label was constructed

Class 1 • 60 flights

41 drone-ship landings, 14 returns to launch site and 5 controlled ocean landings.

Class 0 • 30 flights

Nine explicit failures and 21 records marked None. The negative class includes flights without a confirmed landing.

Predictors

One-hot encode orbit, site, landing pad and booster serial. The resulting matrix has 80 columns. The source table has 26 missing landing-pad entries.

Exploratory visualization

Compare individual flights

Scatter plots compare payload and flight sequence across sites and orbits. Color and marker shape identify the outcome class.

Compare grouped outcomes

For each orbit and year, divide the number of class 1 flights by all flights in the group.

Check the sample size

A 100% rate can represent a single flight. Show the count alongside each orbit rate.

SQL analysis

Select and aggregate

Use DISTINCT for site labels, LIKE for Cape Canaveral records, SUM and AVG for payload, and MIN for the first ground landing.

Filter and rank

Filter drone-ship outcomes by payload or calendar year. Use a maximum-payload subquery and rank counts over a specified interval.

Normalize outcome labels

Trim stray spaces from Mission_Outcome before counting. Retain the separate entry with unclear payload status.

Folium map construction

Locate the sites

Place a circle and label at each distinct coordinate. The nearby Florida labels merge at national scale.

Inspect individual flights

Group overlapping markers with MarkerCluster. Zoom in to distinguish green class 1 flights from red class 0 flights.

Measure a selected distance

Compute a great-circle distance between a launch coordinate and a selected comparison point. The result depends on both coordinates.

Plotly Dash dashboard

All sites

The pie chart groups class 1 flights by site and shows each site’s share of the 24 successful records.

One site

Selecting a site changes the pie chart to success versus other outcomes within that site.

Payload range

A slider filters the payload scatter plot. The vertical axis shows the outcome and color indicates booster version.

Predictive analysis methodology

Partition

Use the 80 encoded features and the binary landing label. Split 90 flights into 72 training flights and 18 test flights with random_state=2.

Tune

Run 10-fold GridSearchCV for logistic regression, SVM, decision tree and K-nearest neighbors. Compare their held-out accuracy.

Verify

Rerun the split with scaling inside each training fold, a fixed tree seed, and valid tree settings. Compare the rerun with the saved notebook.

Results

Flight patterns

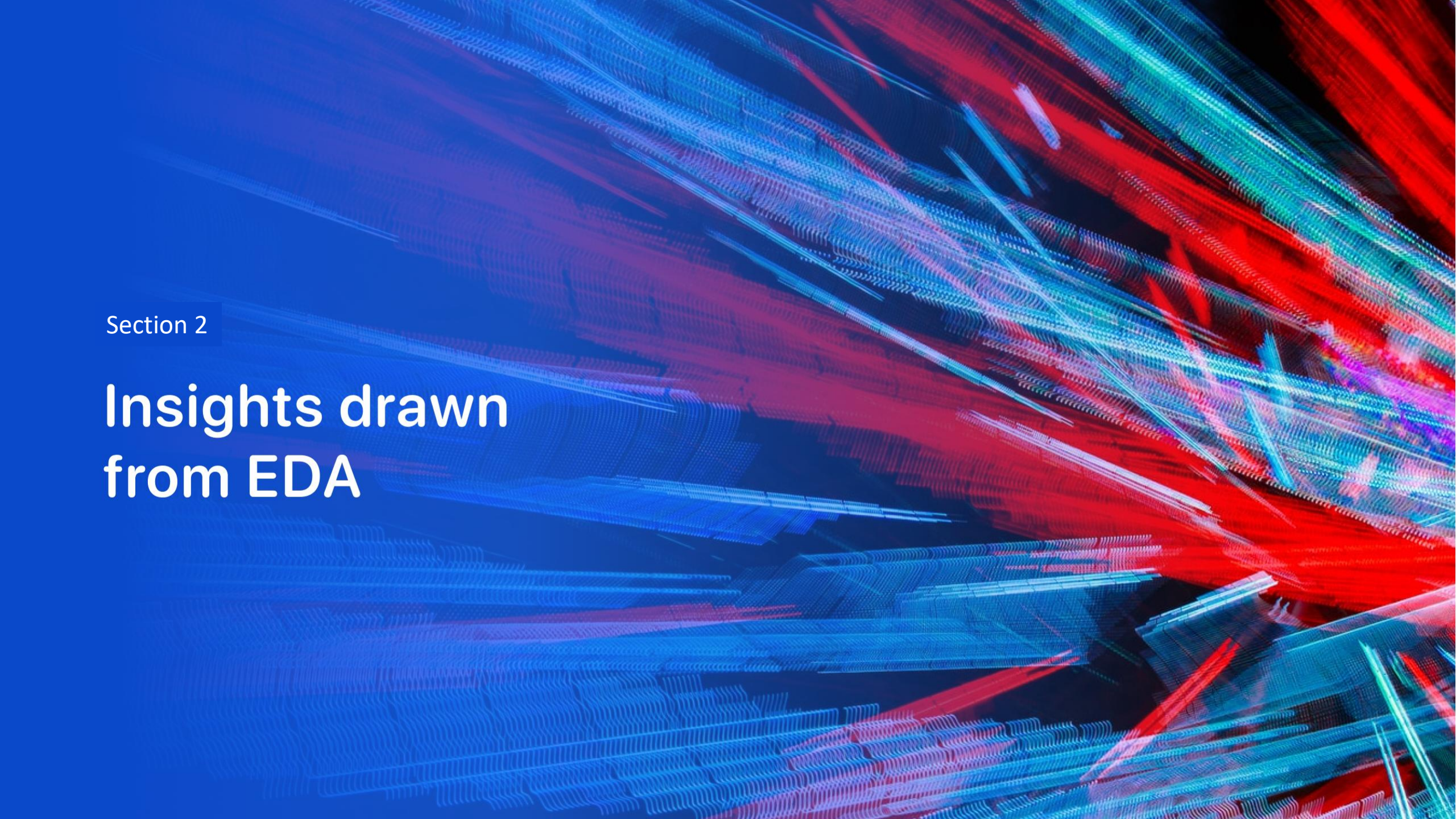
The 90-flight snapshot spans June 2010 to November 2020. Later missions had a higher recorded landing rate than early missions.

Site-level exploration

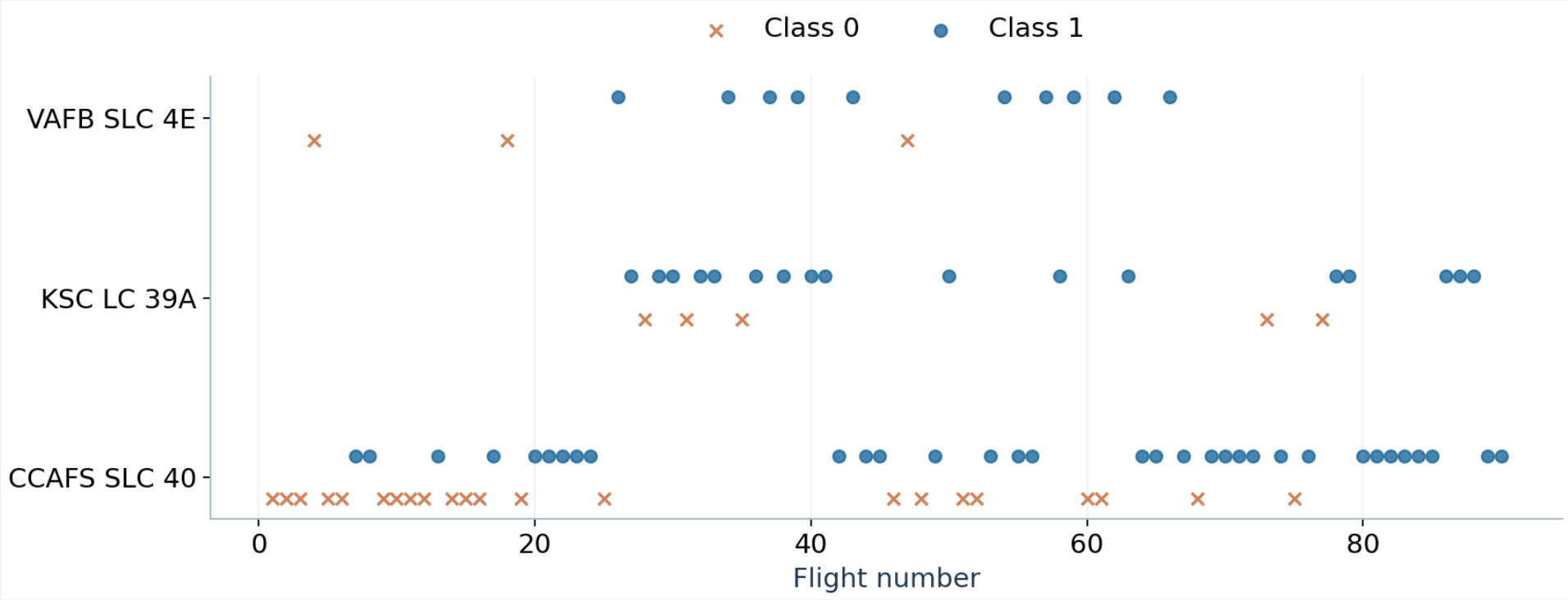
The 56-flight interactive sample places 46 records in Florida and 10 in California.

Held-out prediction

All four classifiers returned the same test accuracy. Their errors matter when a positive prediction informs a reuse assumption.


Section 2
Insights drawn from EDA

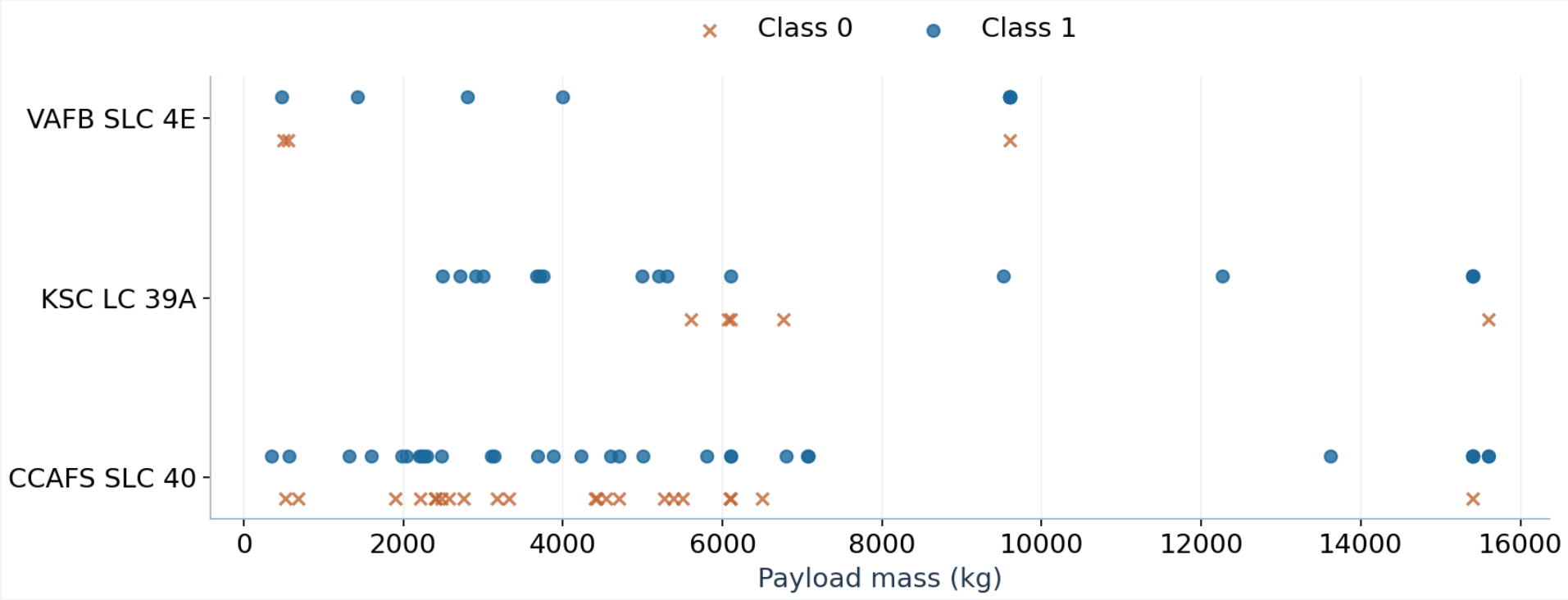
Later flights include more sites



Kennedy first appears at flight 27. Cape Canaveral spans flights 1–90, so raw site comparisons also reflect different operating periods.

Source: exploratory visualization notebook and IBM dataset_part_2.csv

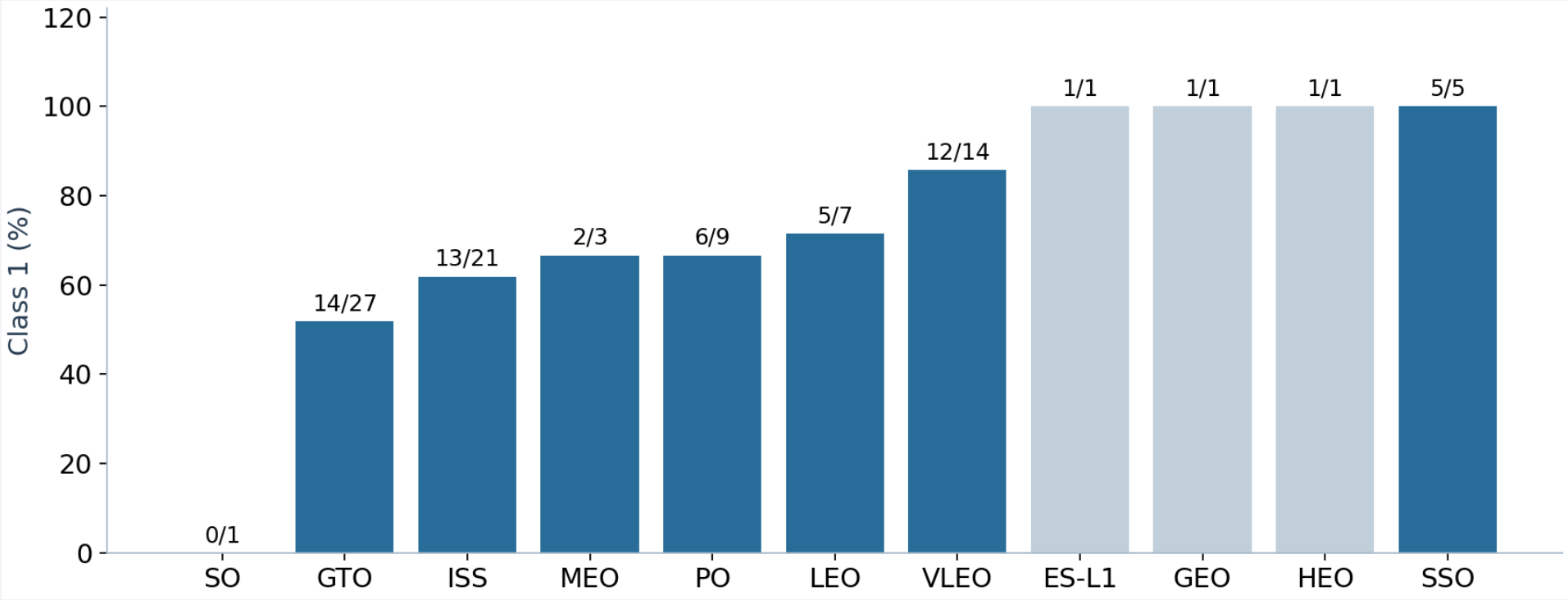
Payload ranges overlap across launch sites



Both Florida sites carried payloads up to 15,600 kg. Vandenberg's largest payload in this sample was 9,600 kg.

Source: exploratory visualization notebook and IBM dataset_part_2.csv

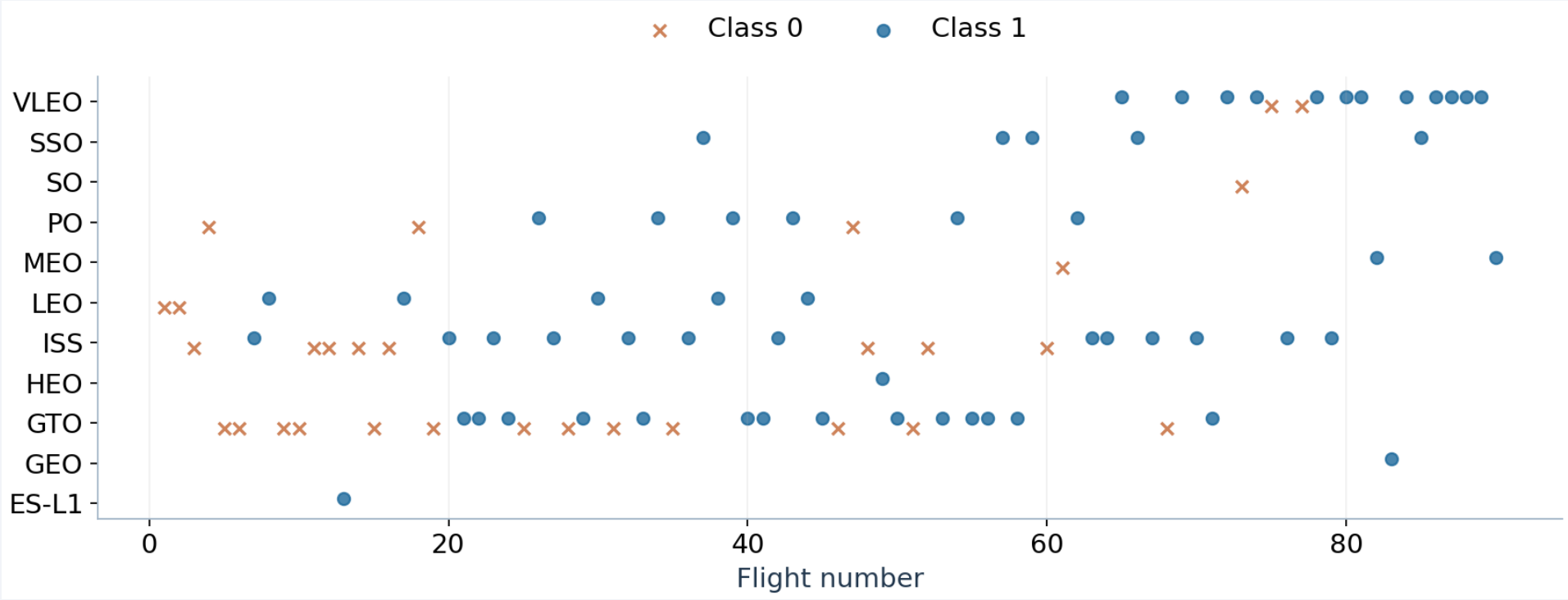
Orbit rates depend on the number of flights



SSO recorded 5/5 positive outcomes and VLEO 12/14. Three other orbits reached 100% on just one flight each. Labels show successes / flights.

Source: exploratory visualization notebook and IBM dataset_part_2.csv

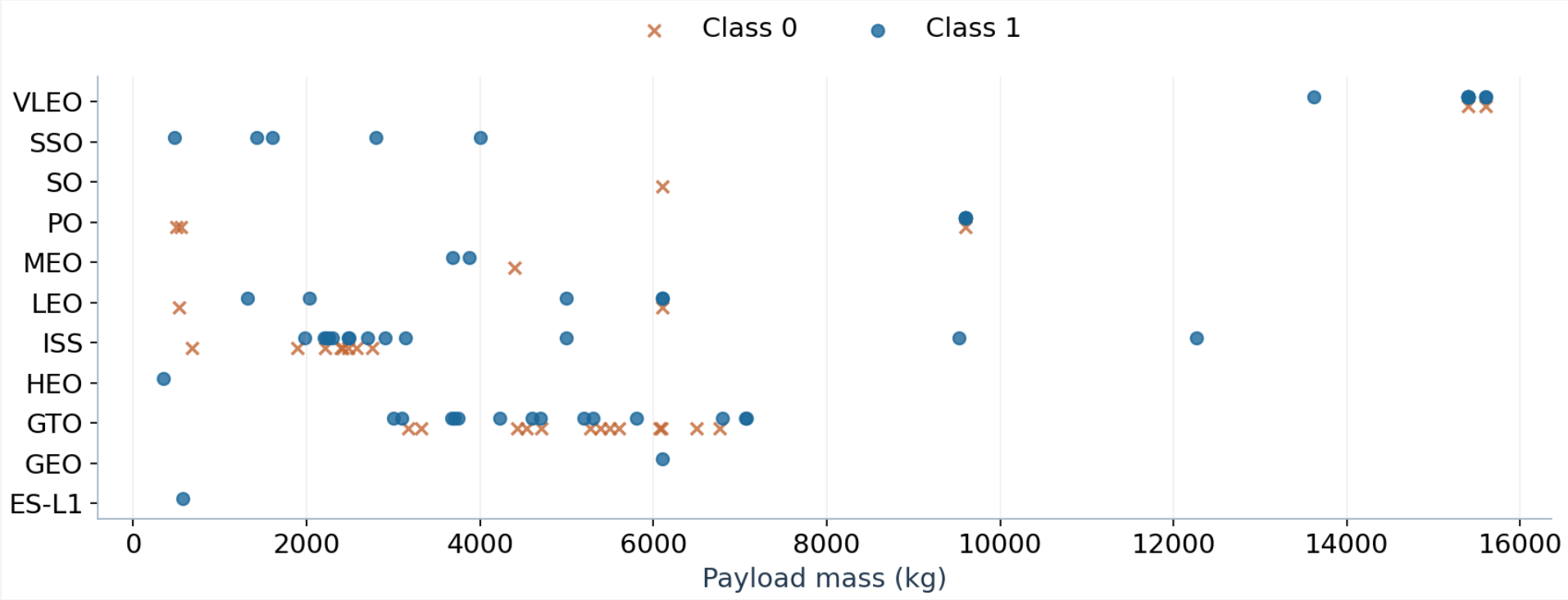
The mix of destination orbits changed over time



VLEO first appeared at flight 65. Comparing its landing rate with earlier orbits also compares different periods of booster operations.

Source: exploratory visualization notebook and IBM dataset_part_2.csv

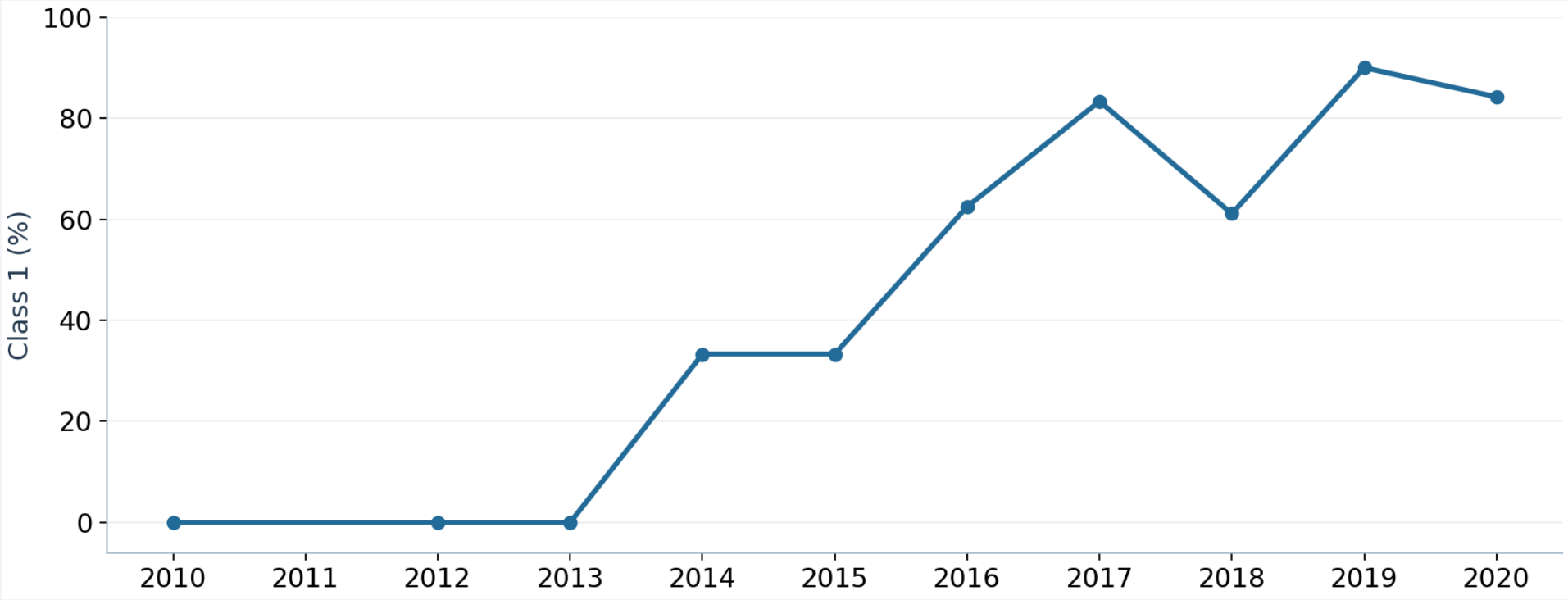
VLEO carried the heaviest payloads in this sample



VLEO payloads ranged from 13,620 to 15,600 kg. Large mass and orbit type overlap, making their separate effects difficult to isolate.

Source: exploratory visualization notebook and IBM dataset_part_2.csv

Annual landing rates rose unevenly



The first five flights across 2010, 2012 and 2013 had no positive outcomes. The rate reached 9/10 in 2019 and 16/19 in 2020. No 2011 flight appears in the sample.

Source: exploratory visualization notebook and IBM dataset_part_2.csv

SQL identified four site labels

Florida

CCAFS LC-40, CCAFS SLC-40 and KSC LC-39A.

California

VAFB SLC-4E.

Interpretation

The SQL extract lists the two Cape Canaveral labels separately. Keep those labels when reproducing its counts.

The first five CCA records came from LC-40

2010

4 June: qualification flight, B0003. 8 December: Dragon demo C1, B0004.

2012

22 May: Dragon demo C2, B0005. 8 October: CRS-1, B0006.

2013

1 March: CRS-2, B0007. A CCA prefix matches Cape Canaveral records. Sorting by date makes the five-row preview reproducible.

NASA CRS payload totaled 45,596 kg

45,596 kg

Total recorded payload mass for SQL rows whose customer is exactly NASA (CRS).

Query

SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)'

Scope

Other NASA customer labels, including COTS, fall outside this filter.

F9 v1.1 payload averaged 2,534.7 kg

2,534.7 kg per flight

Mean payload for records whose booster-version field exactly matches F9 v1.1.

Query

SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1'

Scope

Records with a serial number appended to the version string fall outside this exact-match query.

The first recorded ground-pad success was in 2015

22 December 2015

The earliest launch date in the SQL table with the landing outcome “Success (ground pad)”.

Query

```
SELECT MIN(Date) FROM SPACEXTABLE WHERE Landing_Outcome = 'Success (ground pad)'
```

Interpretation

The date refers specifically to a ground-pad landing in this extract.

Four boosters met the drone-ship payload filter

Payload window

Select successful drone-ship landings with payload strictly above 4,000 kg and below 6,000 kg.

Matching booster versions

F9 FT B1022, F9 FT B1026, F9 FT B1021.2 and F9 FT B1031.2.

Interpretation

Each of these four records combines a specified payload range with a confirmed drone-ship recovery.

Mission success and booster landing are separate outcomes

99 mission successes

Trimming the outcome field merges 98 “Success” records with one record containing a trailing space.

Two other mission records

One in-flight failure and one success with unclear payload status bring the SQL total to 101.

Why this matters

A mission can deliver its payload even when its booster fails to land. Classification uses the first-stage outcome.

Twelve booster records shared the maximum payload

15,600 kg

A subquery finds the maximum payload. The outer query returns all 12 flights tied at that mass.

F9 B5 booster records

B1048.4, B1049.4, B1051.3, B1056.4, B1048.5, B1051.4, B1049.5, B1060.2, B1058.3, B1051.6, B1060.3 and B1049.7.

Interpretation

The suffix identifies the flight of a core. Some core serials therefore appear more than once.

Two drone-ship landing failures occurred in 2015

January 2015

F9 v1.1 B1012 launched from CCAFS LC-40.

April 2015

F9 v1.1 B1015 launched from the same site.

Selection

Both SQL records have the exact landing outcome “Failure (drone ship)”.

Early flights often had no landing attempt

10 records

No attempt was recorded on 10 flights from 4 June 2010 through 20 March 2017, inclusive.

5 records each

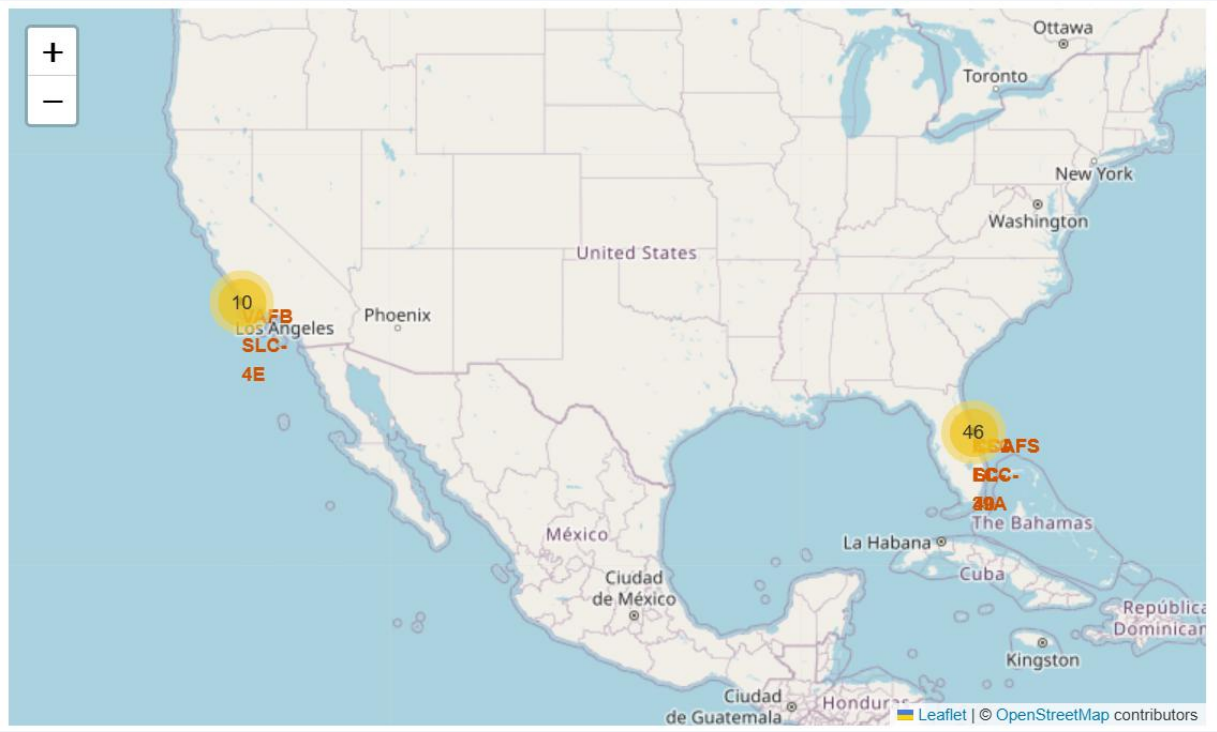
Successful drone-ship landing and failed drone-ship landing tied for second.

Other outcomes

Ground success 3, controlled ocean 3, uncontrolled ocean 2, parachute failure 2, precluded drone ship 1. Total: 31 records.


Section 3
Launch Sites
Proximities Analysis

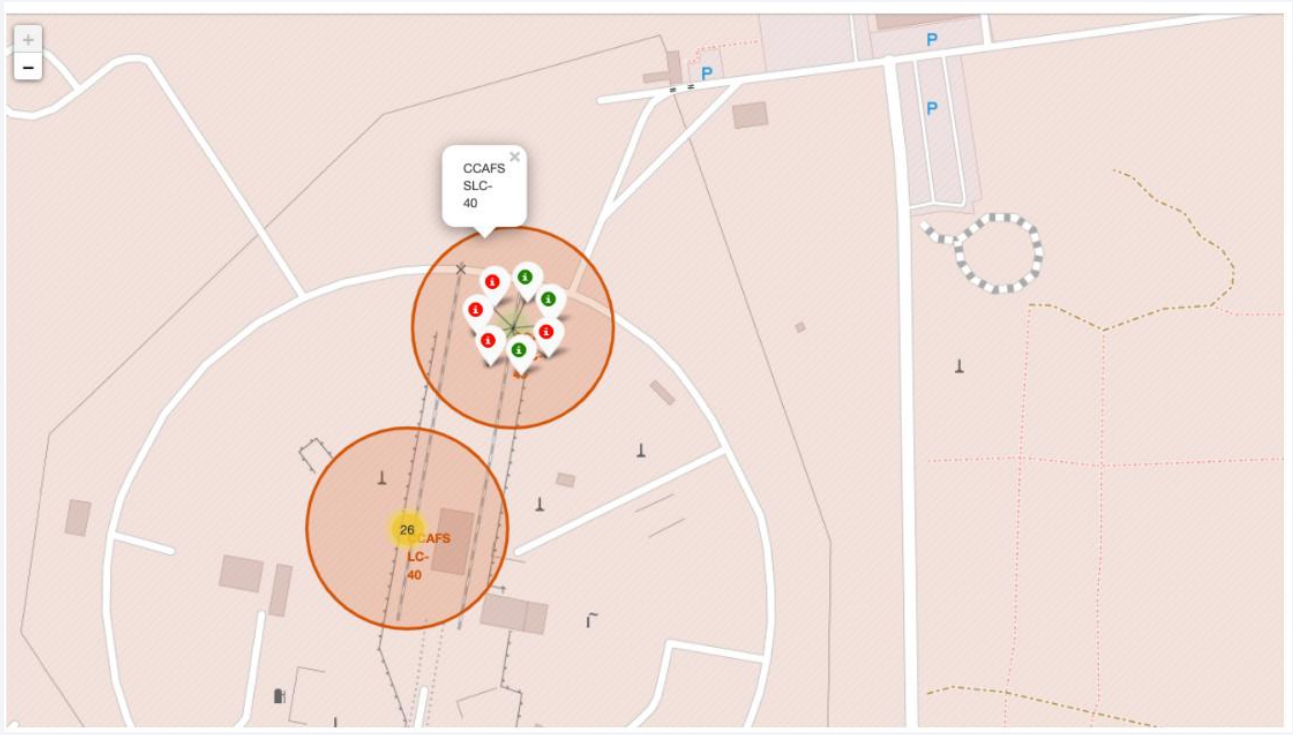
Launch records cluster on two US coasts



The markers represent 56 flights. Forty-six cluster in Florida and ten at Vandenberg in California. Zooming in separates the Florida site labels.

Source: Folium notebook and IBM geographic extract. Map © OpenStreetMap contributors.

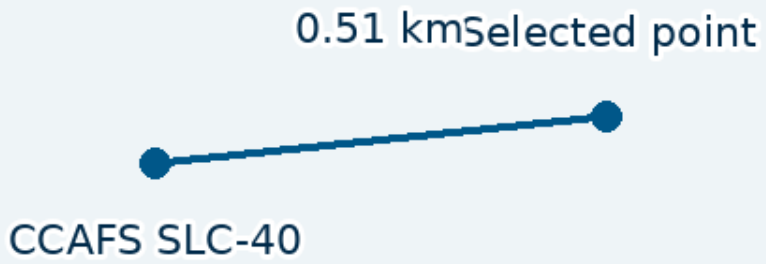
Zooming in reveals mixed outcomes at SLC-40



At CCAFS SLC-40, three of seven records have green markers (class 1) and four have red markers (class 0). The nearby LC-40 cluster contains 26 records.

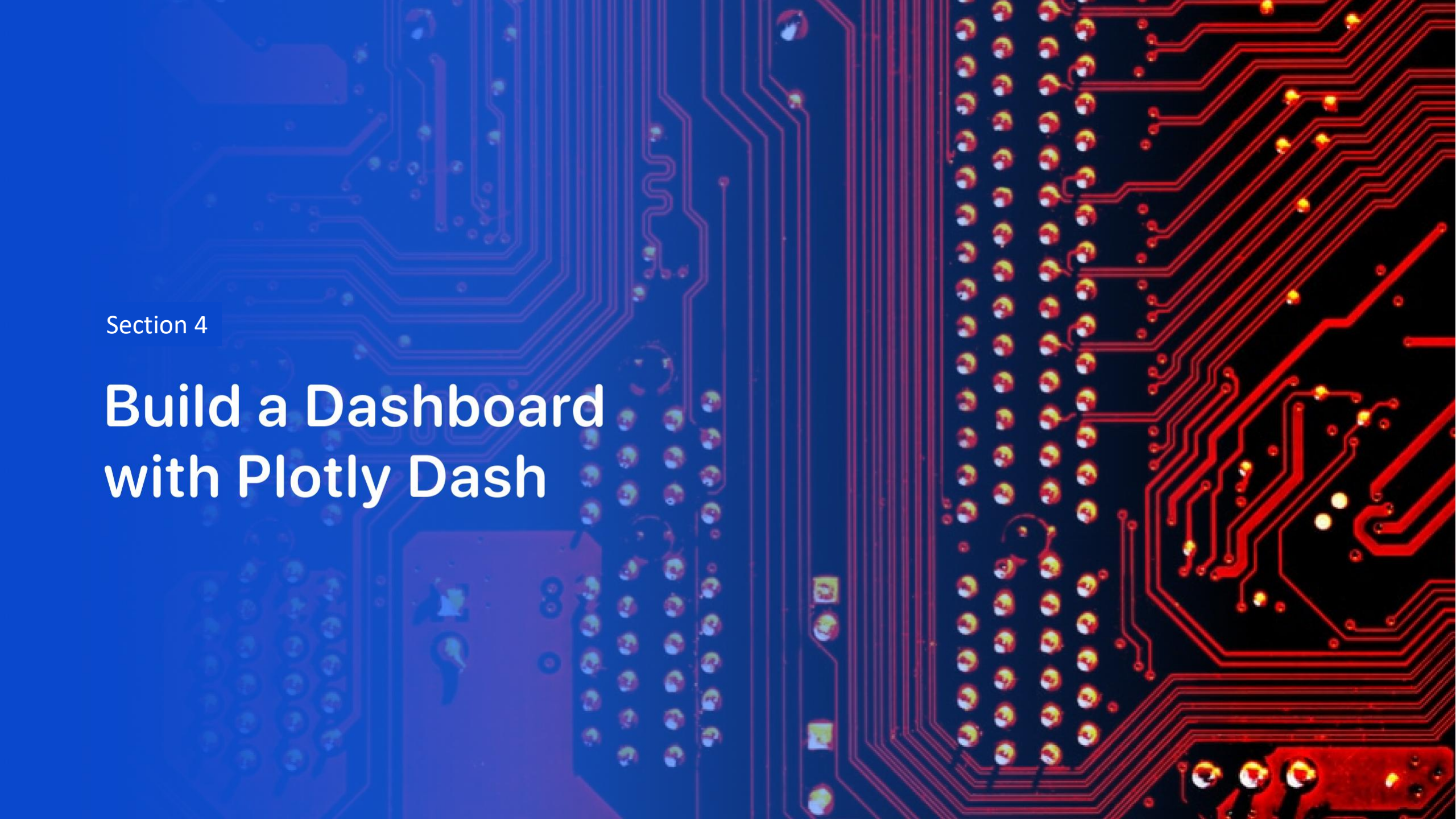
Source: Folium notebook and IBM geographic extract. Map © OpenStreetMap contributors.

Distance to the selected coastline point was 0.51 km

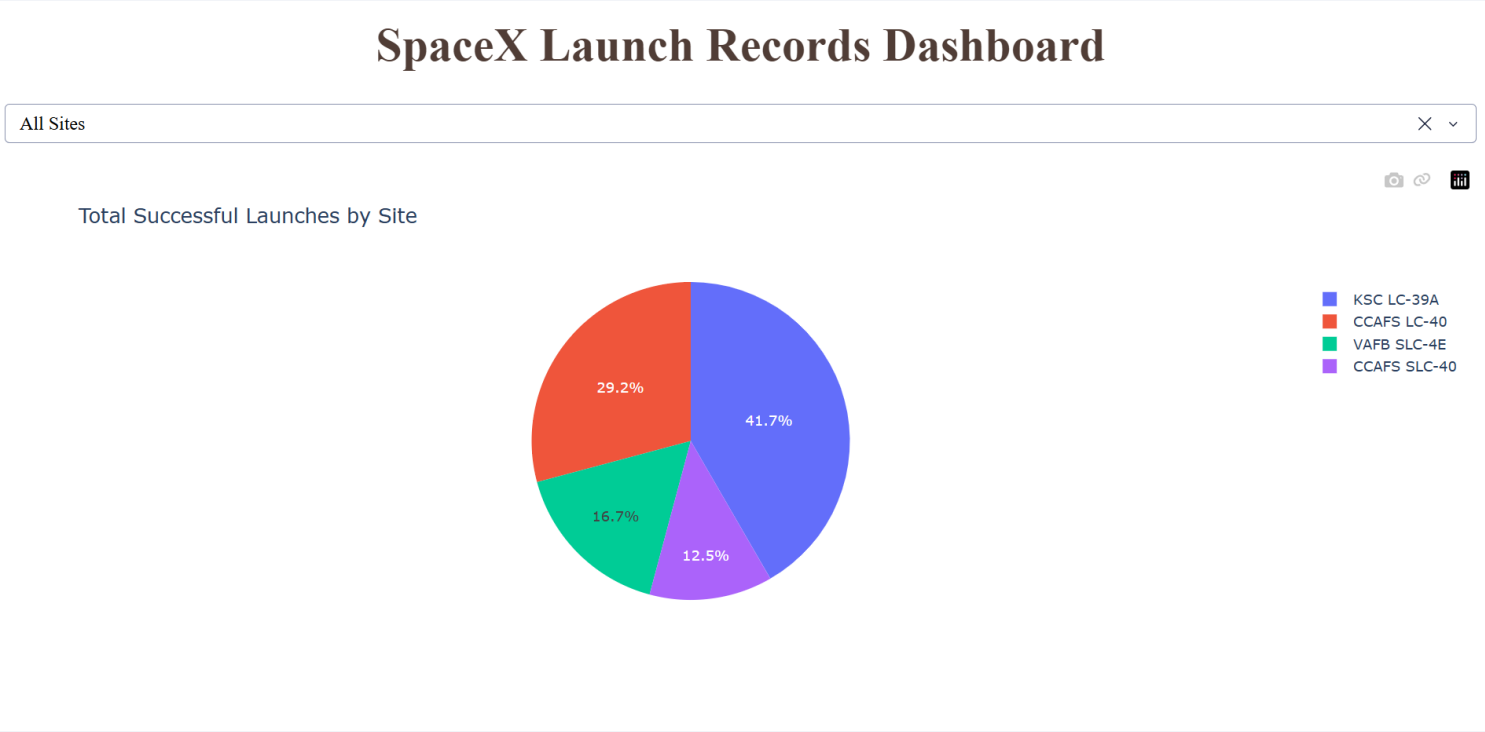


The haversine calculation pairs CCAFS SLC-40 (28.563197°, −80.576820°) with the lab’s selected point (28.563670°, −80.571630°). This measures point-to-point distance.

Coordinate sketch from the Folium notebook. North is up. The chosen point determines the distance.


Section 4
Build a Dashboard with Plotly Dash

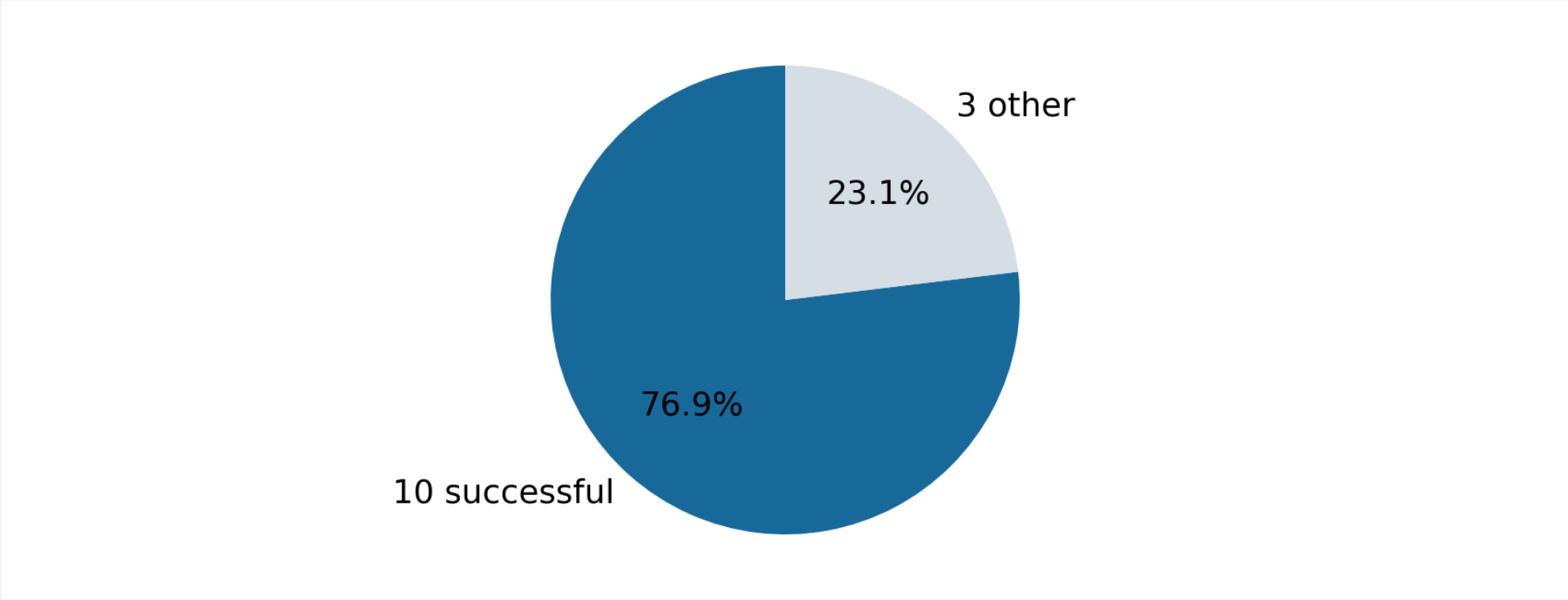
Kennedy accounted for 41.7% of successful flights



Ten of the sample's 24 successful outcomes came from KSC LC-39A. The remaining 14 came from the three other site labels.

Source: original Plotly Dash output, All Sites view.

Kennedy had the highest within-site success rate



At KSC LC-39A, 10 of 13 flights succeeded (76.9%). The corresponding rates were 3/7 at CCAFS SLC-40, 4/10 at VAFB and 7/26 at CCAFS LC-40.

Single-site view reconstructed from the 56-flight geographic extract and checked against the all-sites dashboard.

Similar payloads had different landing outcomes



Both outcomes occur around 3,000–4,000 kg and near 9,600 kg. In this 0–10,000 kg dashboard view, payload mass alone does not separate the classes.

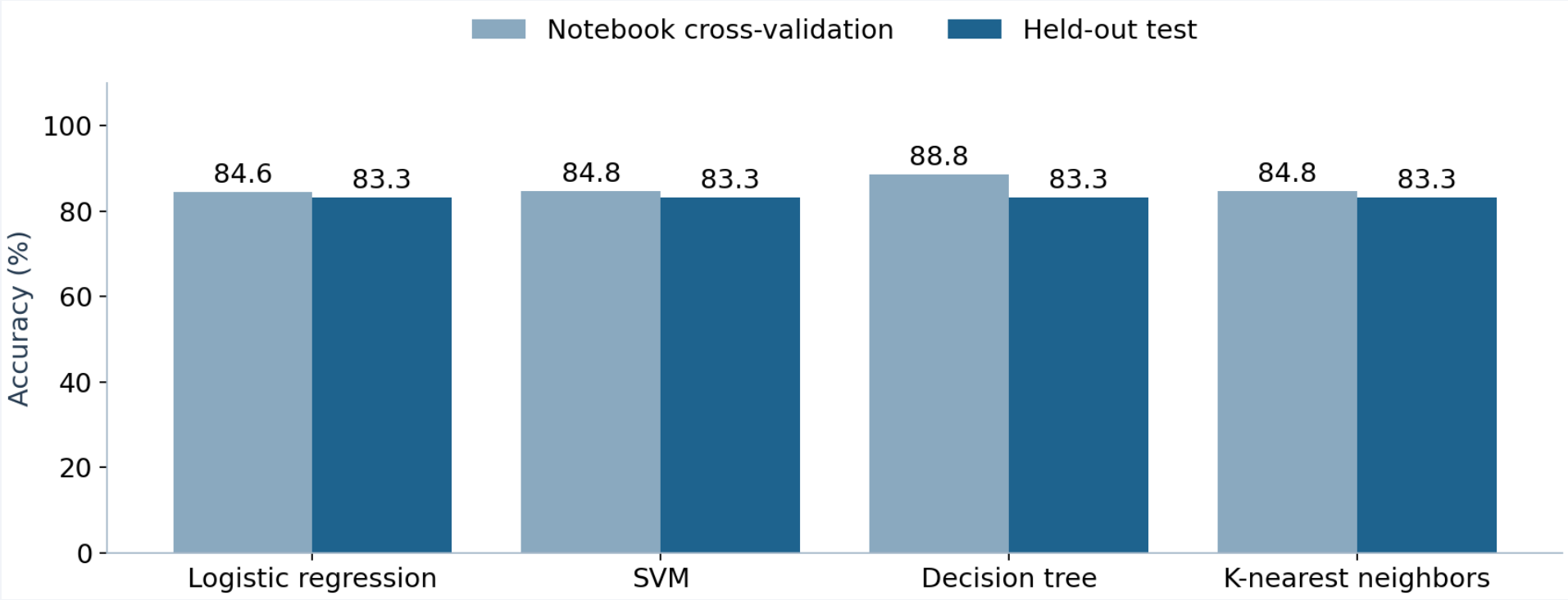
Source: original Plotly Dash output, All Sites and full payload-slider range.

Section 5

Predictive Analysis (Classification)



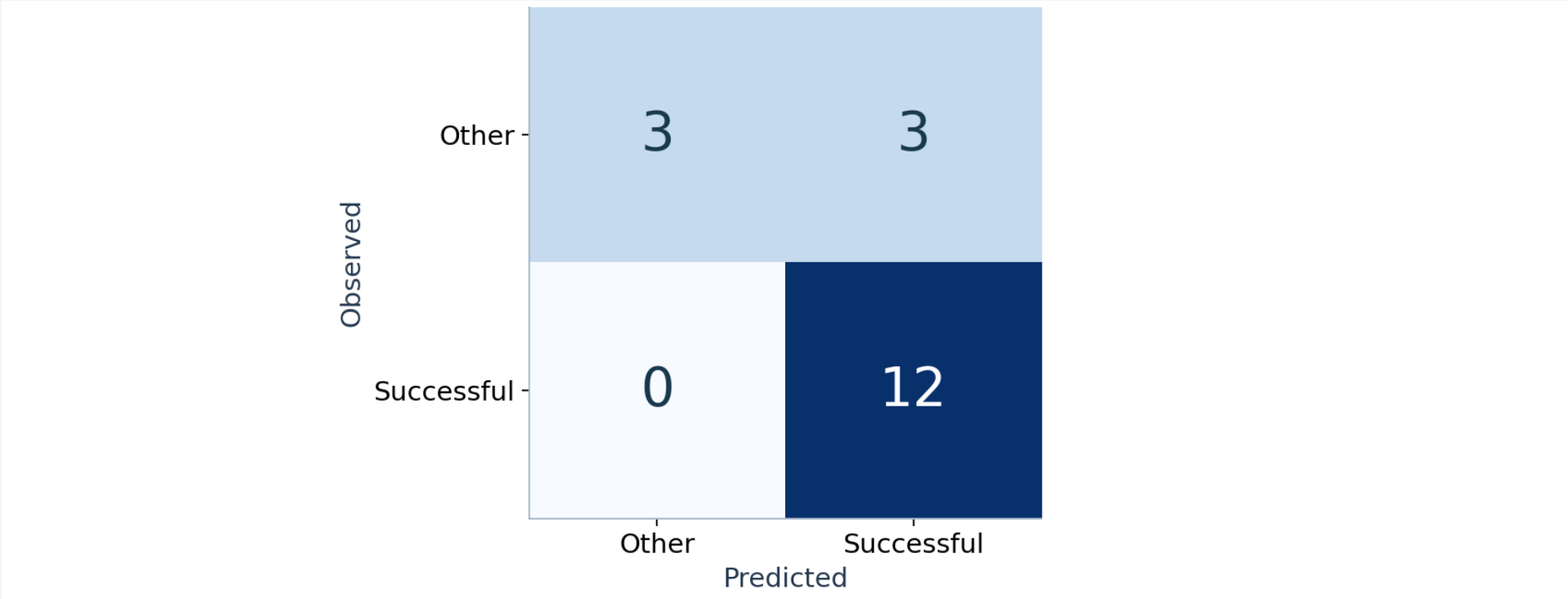
All four classifiers tied on the held-out flights



The tree led notebook cross-validation at 88.75%. On the same 18 test flights, each classifier scored 15/18 (83.3%). The fold-scaled rerun preserved the four-way tie.

Source: saved classification notebook scores. The appendix reports verification-run CV scores.

The models missed three class 0 outcomes



Logistic regression identified 12/12 class 1 flights and 3/6 class 0 flights. A positive prediction was correct 12/15 times (80%). All four verification-run matrices matched.

Source: verification run, 18 held-out flights. Rows are observed classes and columns are predictions.

Conclusions

Recovery became more common

The annual class 1 rate rose from zero in the earliest flights to 16/19 in 2020. Site and orbit comparisons also reflect changes in the mission mix.

The classifiers improve on a simple baseline

Always predicting class 1 gives 12/18 correct. The models give 15/18, but still predict a successful outcome for half the class 0 test flights.

A more realistic next test

Hold out later flights, restrict predictors to values known before launch, and evaluate how often a predicted recovery fails.

Appendix: model verification

Same data and split

Use dataset_part_2.csv for labels and dataset_part_3.csv for features. Use the same 72/18 split and 10-fold training-set grid search.

Changes in the rerun

Fit StandardScaler inside each training fold. Drop invalid max_features=auto, keep sqrt, and fix the tree seed at 2. The original tree grid rejected half its fits.

Verification-run accuracy

Mean CV: logistic regression 84.64%, SVM 82.14%, tree 87.68%, KNN 84.82%. Test: all four 83.33%.

Submission check: https://github.com/Ncontai14/Applied_Data_Science_Capstone_IBM_Coursera

Thank you!



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